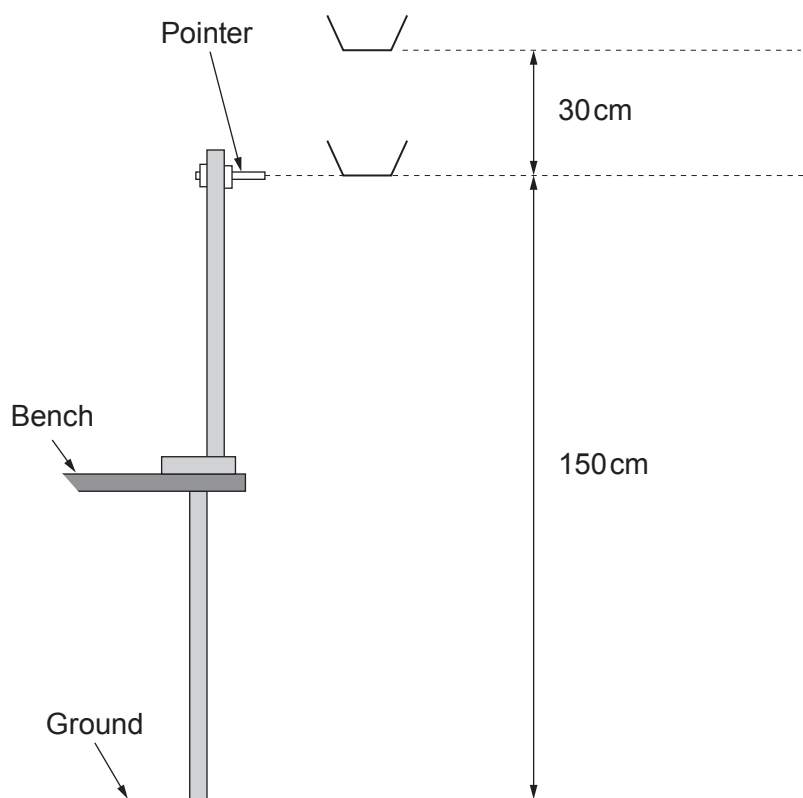


Answer **all** questions.

1. A group of students investigate the terminal speed of a number of cake cases. They set up the following apparatus. They drop the cake cases from rest at a point 30 cm above a pointer. They measure the time taken for the different numbers of cake cases to fall 150 cm, from the pointer to the ground.



- (a) Complete the sentences below by underlining the correct phrase in the bracket. [3]
- (i) In this experiment, the independent variable is the
(number of cake cases / time of fall / height of pointer).
- (ii) In this experiment, the dependent variable is the
(number of cake cases / time of fall / height of pointer).
- (iii) During the first 30 cm of the fall, the cake cases
(speed up / slow down / fall at constant speed).



- (b) Five separate groups in the class carry out the same experiment. Their results and calculated values are shown below.

Distance (cm)	Number of cake cases	Time of fall (s)						Terminal speed (m/s)
		Group A	Group B	Group C	Group D	Group E	Mean	
	0							0
150	1	0.86	0.88	0.91	0.90	0.85	0.88	1.7
150	2	0.72	0.72	0.67	0.65	0.66	0.68	2.2
150	3	0.60	0.59	0.62	0.60	0.61	0.60	2.5
150	4	0.55	0.56	0.55	0.75	0.58		2.7
150	5	0.50	0.55	0.51	0.54	0.50	0.52	2.9

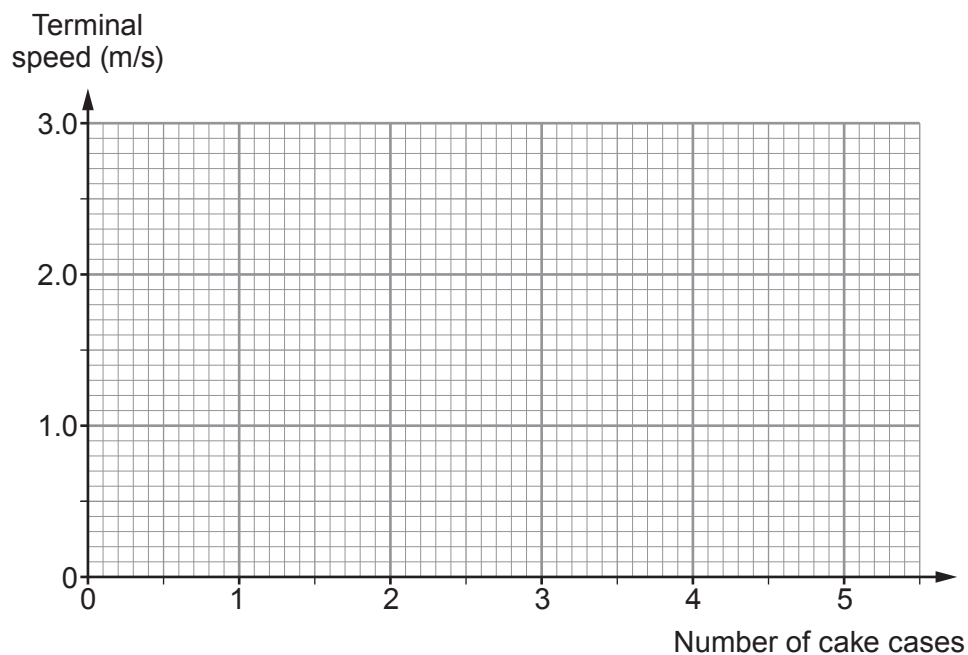
- (i) **Draw a circle** around the anomalous result when **4 cake cases** are dropped. [1]
- (ii) Calculate the mean time for 4 cake cases to fall 150 cm. [2]

Mean time = s



- (c) (i) Use the table below to plot the data on the grid and draw a suitable curve of best fit. [3]

Number of cake cases	Terminal speed (m/s)
0	0
1	1.7
2	2.2
3	2.5
4	2.7
5	2.9



- (ii) Complete the sentence below by underlining the correct phrase in each bracket. [2]

As the number of cake cases increases, the terminal speed (**increases / decreases / stays the same**) at (**an increasing / a decreasing / a constant**) rate.



- (b) One source of inaccuracy in this experiment is measuring the length of the spring. State **two** ways this measurement could be improved. [2]

1.

2.

- (c) The spring has an extension of 8 cm when a mass of 400 g is suspended from it.

- (i) State the weight of the 400 g mass. [100 g weighs 1 N] [1]

Weight = N

- (ii) Use the equation:

$$\text{spring constant} = \frac{\text{weight}}{\text{extension}}$$

to calculate the spring constant for an extension of **8 cm**. [2]

Spring constant = N/cm

- (iii) John says that a mass of 200 g would stretch the spring by 2 cm. Explain whether his claim is correct. [2]

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